

Chapter 8

An LLM-Based Adaptive Learning System for Automated Essay Assessment Aligned with Bloom's Taxonomy



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Abstract This chapter introduces an interactive learning system that uses OpenAI's large language model (LLM) to automatically create and evaluate essay-style questions aligned with Bloom's Taxonomy. The system processes educational PDFs using term frequency-inverse document frequency (TF-IDF)-based keyword extraction (from unigram to trigram) to connect question content to its source, reducing hallucinations. Semantic evaluation is performed with the ada-002 embedding model and cosine similarity to assess student answers. When similarity scores fall below a set threshold, the system provides follow-up questions to support adaptive learning. The platform supports multiple languages and was tested in a two-phase study with university students in Japan and Indonesia. Evaluation using the technology acceptance model (TAM) showed high perceived usefulness (0.991) and ease of use (0.850), with improved results after system updates. Findings also indicate that documents in their original language (English) produce more stable and accurate semantic similarity scores than translated versions, highlighting the limitations of cross-lingual embeddings in AI-assisted educational assessment.

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8.1 Introduction

The integration of artificial intelligence (AI) in higher education is transforming instructional design, assessment, and personalization, with large language models (LLMs) playing a pivotal role due to their ability to understand and generate human-like text [5]. LLMs utilize natural language processing (NLP) to extract key concepts, infer semantic relationships, and generate pedagogically aligned questions. While earlier AI-driven approaches have focused on template-based multiple-choice or short-answer generation [26, 29], they often lack contextual depth and adaptability to diverse cognitive levels, highlighting the unique potential of LLMs for dynamic, context-aware educational tasks.

To address this, our approach utilizes Bloom's taxonomy [7]—a widely adopted hierarchical model of learning objectives—to guide the generation of questions across multiple cognitive levels. At the foundational level, questions test recall (remembering) and understanding; at more advanced levels, they prompt learners to analyze, evaluate, or apply knowledge. Embedding these structures into question generation promotes differentiated instruction, encourages deeper learning, and aligns AI outputs with educational standards.

Beyond question generation, automated assessment is another domain where LLMs can contribute significantly. By generating reference answers directly from instructional materials and comparing them with student responses using semantic similarity (e.g., cosine distance over embedding vectors [35]), AI can perform accurate, scalable, and context-sensitive grading. Unlike traditional keyword-based scoring systems, semantic evaluation accounts for variations in wording and phrasing, offering fairer assessments that recognize conceptual equivalence. This approach not only enhances assessment efficiency but also provides real-time, formative feedback that encourages meta-cognitive reflection and critical thinking.

Our system combines automated question generation and semantic-contextual grading within a single learning platform, using term frequency-inverse document frequency (TF-IDF) representation to extract anchor terms from educational PDFs at unigram to trigram levels, ensuring relevance in context [20]. It also supports multilingual interactions. User acceptance is assessed using the technology acceptance model (TAM) [10], focusing on perceived usefulness (PU) and ease of use through (PEU), evaluated with Japanese and Indonesian university students, highlighting adaptive interaction and pedagogical alignment.

Our study presents the following contributions:

- A novel system that generates Bloom's Taxonomy-aligned essay questions using OpenAI's LLM, anchored with local and global context from TF-IDF-based keyword extraction.
- Integration of semantic similarity scoring, based on TF-IDF, using OpenAI's text embedding ada-002 model to compare student answers with reference responses.
- An adaptive questioning mechanism that provides follow-up prompts if students' answers do not meet cosine similarity thresholds.

- Multilingual implementation and testing in two distinct academic environments (Japan and Indonesia).
- Empirical validation using TAM, highlighting system usability, perceived benefit, and cross-cultural applicability.

These contributions position our work at the intersection of educational technology, natural language processing, and pedagogical design, making it one of the first systems to offer an end-to-end solution for context-aware question generation and semantic grading within a multilingual, web-based learning platform.

8.2 Related Work

This section reviews the state-of-the-art in AI-driven educational technologies, focusing on five thematic areas: (1) Automated essay scoring, (2) Embeddings and text similarity in educational NLP, and (3) TAM in education.

8.2.1 *Automated Essay Scoring*

Cosine similarity remains an important method in text analysis and automated essay scoring (AES), valued for its ability to measure the similarity between student essays and reference texts regardless of word order. While early AES systems relied on hand-crafted features and traditional classifiers [4, 34], modern systems like ETS's e-rater and IntelliMetric utilize NLP and latent semantic analysis to assess coherence and argumentation [12]. Recent research continues to apply cosine similarity effectively for comparing essays to manually graded references, supporting more nuanced and scalable assessment in educational settings.

Cosine similarity has gained prominence in AES due to its ability to evaluate student-generated responses against reference answers semantically. Fauzi et al. [14] used n-gram and cosine similarity to evaluate short essays, while Lahitani et al. [19] implemented cosine-based scoring in Indonesian educational contexts. These models, however, struggle with understanding syntactic variation and contextual nuance—gaps addressed by transformer-based models such as BERT [11].

Overall, cosine similarity, whether using the bag-of-words method or TF-IDF, has proven effective in developing fast and efficient automatic essay scoring systems. However, further development, such as applying n-grams and considering word order, can enhance accuracy in essay assessment, particularly in contexts that require a deeper understanding of sentence structure and meaning.

The alignment of question generation with Bloom's Taxonomy has been explored to enhance instructional design. Alammery et al. [1] proposed a modified TF-IDF method to generate Arabic-language questions categorized by Bloom's levels. Melevsko [25] and Scaria et al. [32] demonstrated that taxonomically informed question generation leads to better learning outcomes.

Recent work by Muñoz et al. [26] introduced grammar-based automatic question generation systems for compilers education, though not Bloom-aligned. Generative LLMs, like GPT-3, have begun to bridge this gap by adapting embeddings and prompt engineering to organize questions according to cognitive level [23, 39].

8.2.2 *Embeddings and Text Similarity in Educational NLP*

Embeddings, such as OpenAI's ada-002, are powerful numerical representations of text that capture semantic meaning across 1536 dimensions. This enables systems to understand contextual relationships between words or phrases with high precision. The ada-002 model is not only effective across various NLP tasks but also optimized for cost-efficiency, making it suitable for resource-constrained educational settings [21].

In automatic assessment and interactive question answering systems, embeddings like ada-002 support semantic-level evaluation, allowing student answers to be judged even when phrased differently from the reference. This flexibility enhances accuracy in grading and enables more responsive feedback. Systems like those described by Serrano et al. [33] and Patel et al. [28] show how embeddings improve retrieval-augmented generation, enabling adaptive and personalized learning experiences.

On the other hand, cosine similarity utilizes TF-IDF, a statistical method for evaluating a word's relevance in a document compared to a set of other documents. TF-IDF determines a word's significance using two components [30]. The first is the term frequency (TF) which measures how frequently a word appears in a specific document relative to the total number of words in that document. The second is the inverse document frequency (IDF) which assesses the significance of a word throughout the entire collection of documents, assigning greater importance to words that occur less frequently in other documents.

The basic formula for TF-IDF is:

$$\text{TF-IDF}(t, d, D) = \frac{f_{t,d}}{\sum_{t' \in d} f_{t',d}} \times \log \left(\frac{N}{|\{d \in D : t \in d\}|} \right) \quad (8.1)$$

Where:

t: term

d: document

D: collection of documents (corpus)

$f_{t,d}$: frequency of term t in document d

N : total number of document in corpus D

$|\{d \in D : t \in d\}|$: number of documents where the term t appears

TF-IDF is commonly used in text analysis, such as keyword extraction, text classification, and document indexing. In the context of education, TF-IDF enables the assessment of word relevance in teaching materials, aids in automating question generation, and supports technology-based learning. For example, Alammary developed a modified TF-IDF method to extract features from Arabic language assessment questions according to Bloom's taxonomy, which improved classification accuracy and relevance [1].

Meanwhile, Qaiser and Ali utilized TF-IDF to identify relevant keywords across different domains from documents. Their research results demonstrated the effectiveness of TF-IDF in text data analysis for various applications, such as document categorization and automatic tagging [30]. Cosine similarity remains the metric of choice for semantic scoring in education due to its efficiency and alignment with human judgments [30, 36]. Combining embeddings with cosine similarity allows systems to move beyond keyword overlap and better approximate instructions [31].

The global use of AI in education necessitates robust multilingual support, yet performance often drops in low-resource languages. Studies by Pesovski et al. [29] and Wang et al. [38] highlight that models like GPT perform optimally in English but less effectively in other languages. While methods such as training on multilingual corpora [8], cross-lingual embeddings [3], and back-translation [24] offer solutions, they come with added latency and computational costs. This motivates the use of lightweight alternatives like our TF-IDF-based approach, which anchors keyword contexts locally.

8.2.3 *Technology Acceptance Model Evaluation*

TAM provides a framework to understand how users adopt educational technologies by focusing on two main factors: perceived usefulness (PU) and perceived ease of use (PEU). PU assesses whether users believe the technology improves their task performance, while PEU evaluates how easily the technology can be used without frustration. These factors are particularly important for automatic assessment systems that use TF-IDF and Cosine Similarity, where user acceptance depends on the system's perceived benefits and user-friendliness.

TAM has been effectively applied in evaluating a range of technologies, including learning management systems [13], mobile learning apps [16], and AI-based tutoring tools [2]. Studies by Chen et al. [5] and Dahri et al. [9] highlight that adaptability and clarity—aligned with PU and PEU—are critical to positive adoption outcomes. By applying TAM, developers can enhance both the functionality and user acceptance of educational technologies [10].

8.3 Method

8.3.1 Overall Methodology

Our research methodology includes a LLM-based Learning System development. The web system is developed by utilizing the OpenAI API and a prompting technique based on a few-shot mechanism. The core of the system lies in the context extraction and anchoring. Context extraction is performed by implementing the TF-IDF formula and extracting the top-10 uni-, bi-, and trigrams from an uploaded document (details in Sects. 8.3.3 and 8.3.4).

Each extracted context will be further associated with a Bloom's level and incorporated into the OpenAI prompt to generate a series of questions. The complete research methodology can be seen in Fig. 8.1. After the system is populated with a series of questions, students can engage in question-and-answer interactions. Each interaction will be evaluated with an automated TF-IDF scoring system, on a 0-100 scale.

To test the system, we conduct a two-step TAM evaluation. The first is using a general text with the topic of Generative AI taken from [27]. The second step is an enhancement phase, which was tested in a real classroom situation, using two chapters from a textbook [15] as the document source.

8.3.2 Technological Architecture

Our LLM-based Learning System integrates state-of-the-art technologies to create a practical and interactive educational platform. It enhances the teaching and learning experience by leveraging web-based architecture, generative AI, and secure data management practices [17]. Figure 8.2 shows the technological architecture involved in developing and implementing the LLM-based Learning System.

Lecturers can upload PDF documents through the Laravel front end, as depicted in Fig. 8.2 of the system flow diagram. The backend processes these uploads, securely storing PDF data in server storage and logging metadata such as uploader information and timestamps in the MySQL database. Students access educational materials and assessments through the same Laravel-based front end. The system uses session management and authentication mechanisms to ensure secure access. It facilitates API-driven communication between the front and back ends for data retrieval and performance tracking (see Fig. 8.2, the figure is initially published in [37]).

Metadata from uploaded PDFs is stored in a MySQL database. This database also manages basic file information and retrieves data for generating assessments and tracking student performance. The stored data is used to generate questions and compile performance metrics, which are then presented to students.

8.3.3 Context Extraction

Document processing begins by applying TF-IDF vectorization to the uploaded text, covering unigrams to trigrams. In this study, a machine learning textbook sub-chapter “Regularized Linear Models”, is used as the baseline document, aligning with the classroom context [15]. Observations indicate that tetragrams produce very low word frequencies, nearing zero, and do not meaningfully improve the model’s performance (see Fig. 8.3).

Next, based on the TF-IDF results, the terms are sorted by cosine similarity values calculated across all uni-, bi-, and trigrams. The top-10 highest values, reflecting the most relevant words or phrases in the document, are selected. These top-10 terms are then incorporated into the prompt, as described in Table 8.1, to provide context when generating questions and answers for each document page. This method ensures that the generated questions are pertinent to the document’s content and uphold strong semantic ties with the uploaded text.

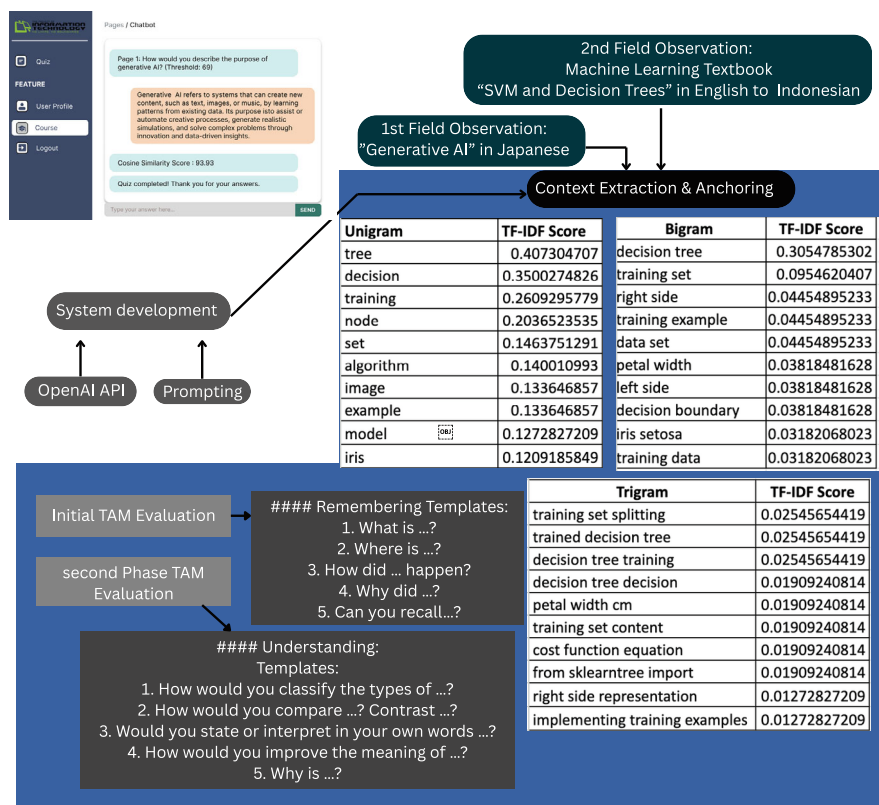


Fig. 8.1 Overall methodology

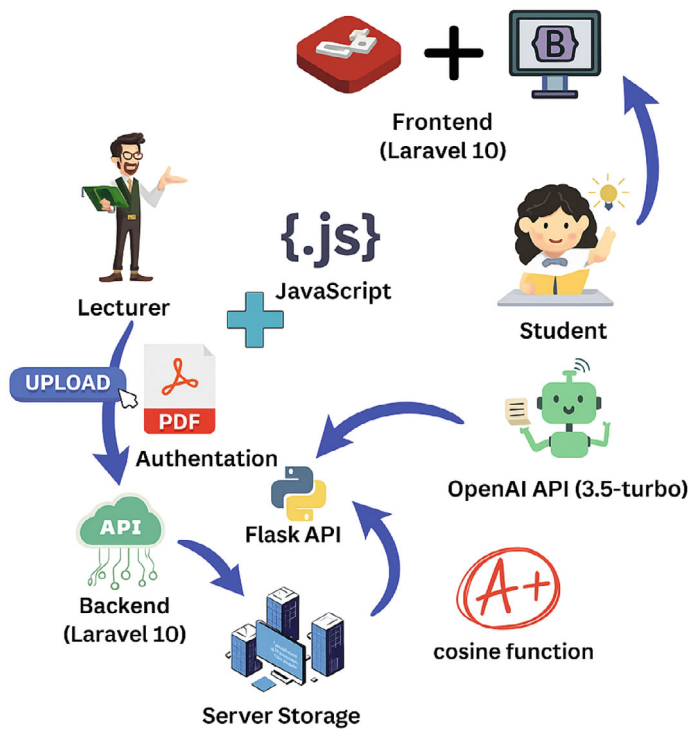


Fig. 8.2 Technological architecture in the developed LLM-based learning system

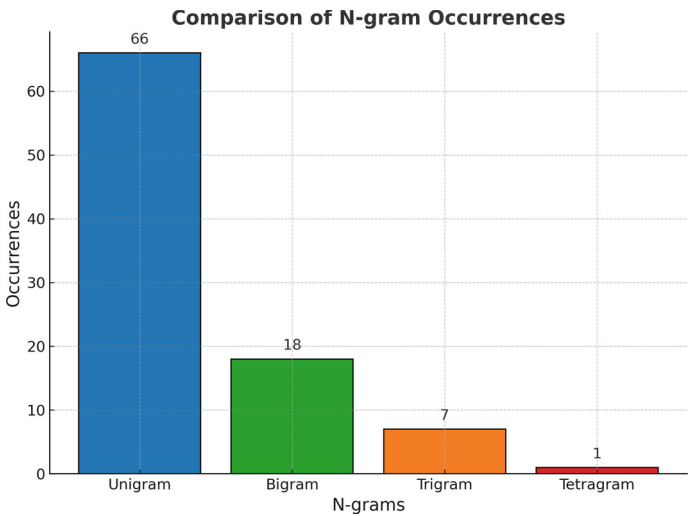


Fig. 8.3 Number of occurrences for each n-gram in the observed text book

Table 8.1 Prompt for constructing contextual questions

Role: You are a bot designed to assist the lecturer. Task: Your task is to construct four critical questions in <code>""" + language + """</code> to test students' knowledge according to the noun phrases in the text. First, translate the user's or text into language <code>""" + language + """</code>. The questions should be structured based on the <code>""" + context anchors + """</code> (see section 8.3.4 for detail) you will use a question template based on Bloom's Taxonomy to generate questions that match the desired level in the following templates make sure you locate the answer from a whole sentence from where the noun phrase is extracted. Few shots: The remembering and understanding question templates are given as few-shot examples in the prompt following the guidance in [18] (due to page limitation, we do not expand the templates in this manuscript). Response: Finally, create two understanding questions and two remembering questions. The response must be in JSON format with question, answer, and category elements.
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8.3.4 Context Anchoring

When generating questions from a document, especially in cases of language mismatch (such as an English source document translated into Indonesian for question-answering), the system first calculates TF-IDF values from the translated version. It then selects the top-10 uni-, bi-, and trigrams from the original language based on the highest cosine similarity scores. These n-grams form the foundation for context anchoring [22], guiding the LLM to focus on the most relevant content. This approach helps prevent the LLMs from generating outdated or semantically unrelated responses by limiting the output to contextually grounded terms.

Three different context anchoring methods were evaluated to identify the most effective approach for the model.

1. In the first method, we examine the anchors used to generate the prompt by selecting the best TF-IDF weighted 'nouns' keywords from the entire document. Each page thus receives a fixed set of global contexts. These nouns—comprising 10 uni-, 10 bi-, and 10 trigrams—are embedded along with the full document text, and cosine similarity is used to find the top-10 best contexts.
2. In the second method, a dynamic anchoring approach is used, with each page assigned a local set of 'nouns' keywords based on the computed TF-IDF of that page. The top-10 uni-, bi-, and trigrams from each page are incorporated into the page's text, and cosine similarity identifies the best top-10 contexts.
3. In the third method, no specific context anchoring is applied. Instead, the GenAI decides which context to use in the generated questions. However, the prompt includes the page text and Bloom's taxonomy templates.

For the first and second methods, the top-10 extracted (unified) uni-, bi, and trigrams are then used as the context anchors in the prompt, as shown in Table 8.1. These anchors are then used to generate the questions, along with the available Bloom's taxonomy templates. We assume that the anchors will provide a certain 'closeness' of important contexts in the course materials. Thus, each course material (an uploaded document) should have some semantic space acting as 'anchors' where all words are 'connected'. The goal of this anchoring method is to ensure that the generated questions and answers are closely tied to the document's content, thereby minimizing OpenAI's hallucinations (inaccurate or irrelevant responses).

8.3.5 *Prompting and Adaptive Mechanism*

The system prompt in Table 8.1 illustrates context anchoring by utilizing nouns extracted from the document to generate relevant questions. These nouns represent core information or key concepts. The noun list is included in the prompt to ensure a strong relevance between the generated questions and the learning topic. Additionally, the prompt incorporates Bloom's Taxonomy templates for Level 1 (Remembering) and Level 2 (Understanding), where Remembering emphasizes basic questions (e.g., definitions and facts) and Understanding evaluates students' ability to interpret and compare concepts.

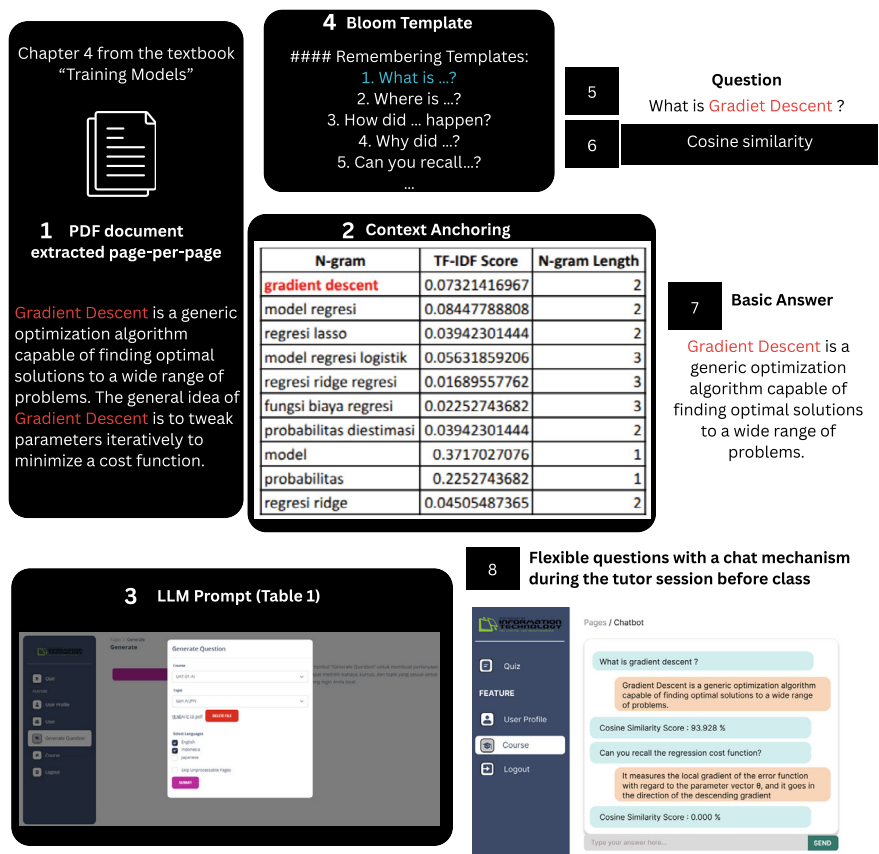
This anchoring directs the system to generate adaptive questions based on local context (key terms from a particular page) while considering global context (key terms from the whole document). The process ensures the relevance of questions and enhances understanding by broadening coverage based on key terms page by page. After generation, nouns from each question are extracted and stored in a list of important nouns, *i.e.* the context anchors for further testing. This method produces varied - relevant questions - that link local and global contexts, thereby fostering a deeper and adaptive learning experience.

8.4 Experiment and Discussion

8.4.1 *System Simulation*

Figure 8.4 illustrates the system's workflow, beginning with uploading a PDF document that contains instructional material, such as Chapter 4, "Training Models," from a referenced textbook [15]. The document is parsed page by page to extract the text content. A highlighted section introduces the concept of "gradient descent," which serves as an anchor for generating learning interactions.

After extraction, the system uses TF-IDF analysis for context anchoring to identify important uni-, bi-, and trigrams. The phrase "gradient descent" is identified as a



material. The system enables multi-turn questioning to promote deeper understanding, with each student's response evaluated using cosine similarity, a semantic measure that compares the student's answer to the source text. High similarity scores, such as 93.928%, indicate strong alignment, while low scores (e.g., 0.000%) prompt adaptive questioning through repetition or rephrasing.

The bottom part of the image shows the full web-based tutor interface. A sidebar provides key features like Quiz, User Profile, Generate Question, and Course Management. Using the "Generate Question" panel, users can upload a PDF, select a course, and a language (English, Indonesian, or Japanese). This starts the automated question generation process, with the chatbot-like interface supporting real-time interaction, answer evaluation, and feedback, creating a smooth and integrated learning experience.

8.4.2 Context Extraction Accuracy

The observation for context extraction and anchoring started with processing course material in the form of a PDF file in English. The material for this evaluation covers Regularized Linear Models, a chapter from the used textbook [15], which spans 20 pages. The English document was translated into Bahasa Indonesia using OpenAI. Additionally, the original English document was also used without translation for further evaluation.

To assess semantic relevance, two remembering and two understanding questions were created for each page, with extra questions generated based on scores from low to high. Using the ada-002 model, embeddings were computed from questions, answers, and page text for both English and Indonesian versions. Cosine similarity was then employed to measure the semantic closeness between questions, answers, and their corresponding page content, allowing a bilingual evaluation of contextual alignment.

The analysis results showed that in the Indonesian document, mostly the cosine similarity values for questions were above 0.85, indicating a high semantic relevance with the content of the translated document. The highest value reached 0.89, indicating a very high semantic alignment. However, the lowest similarity value was 0.76, which was caused by a language mismatch. In some cases, questions were written in English, such as *What happens when the **hyperparameter** alpha value is set to 0 in Ridge Regression?* The word 'hyperparameter' should be written as **hiperparameter** when translated into Indonesian.

This mismatch resulted in a lower cosine similarity values because text embedding could not fully capture cross-lingual semantic relevance (Fig. 8.5). Similar issues were also found in the answers, where answers that were not entirely written in the same language as the translated document tended to have lower similarity values. Nevertheless, the Pearson correlation between question similarity and answer similarity showed a value of 0.76, indicating a fairly strong relationship. This means that

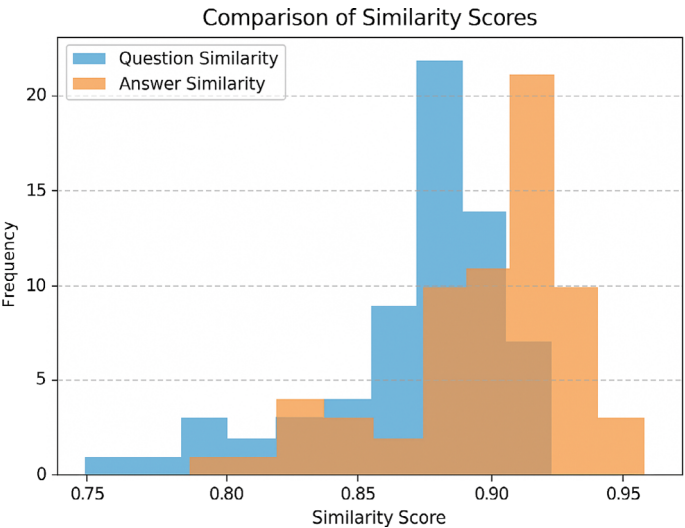


Fig. 8.5 Performance test results of OpenAI with questions and answers in Indonesian

when the question similarity score is high, the answer similarity score also tends to be high, suggesting that both follow a consistent relevance pattern with the document.

The original English document showed higher and more stable semantic similarity scores, mostly above 0.85 and peaking at 0.92, compared to the translated version. This indicates a stronger alignment between questions, answers, and source text (Fig. 8.6). Despite this, both versions had a similar Pearson correlation (0.76) between question and answer similarity, indicating consistent relational patterns across languages. These results emphasize the benefit of language consistency in improving embedding-based semantic evaluations.

8.4.3 Context Anchoring Comparison

Our context anchoring observation shows that—comparing between the global (full document) and local (page-level) contexts—the TF-IDF-based anchoring (left graph in Fig. 8.7) yields stable cosine similarity scores in both scenarios. Dynamic page-level anchoring (middle graph in Fig. 8.7) demonstrates greater local consistency but presents a significant decline in global relevance. Unanchored prompts (right graph in Fig. 8.7) reveal inconsistent cosine similarity scores with broader dispersion. Therefore, the global anchoring controlled by the TF-IDF weights has been shown to be the most stable and precise approach for generating (prompting) contextually relevant questions, as described in Table 8.1.

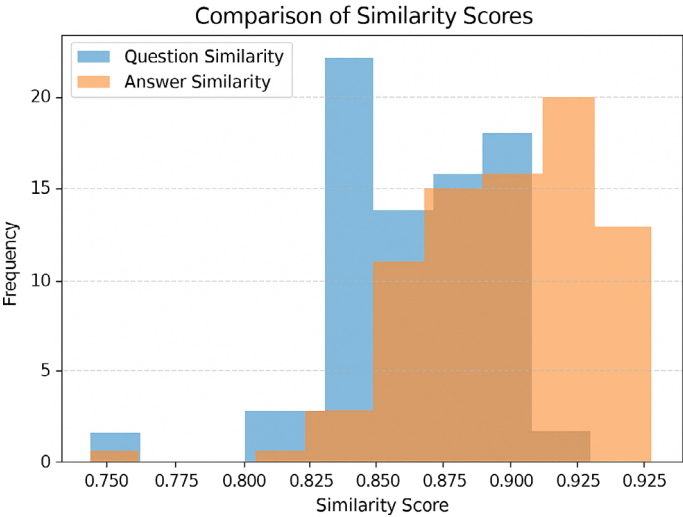


Fig. 8.6 Performance test results of OpenAI with questions and answers in English

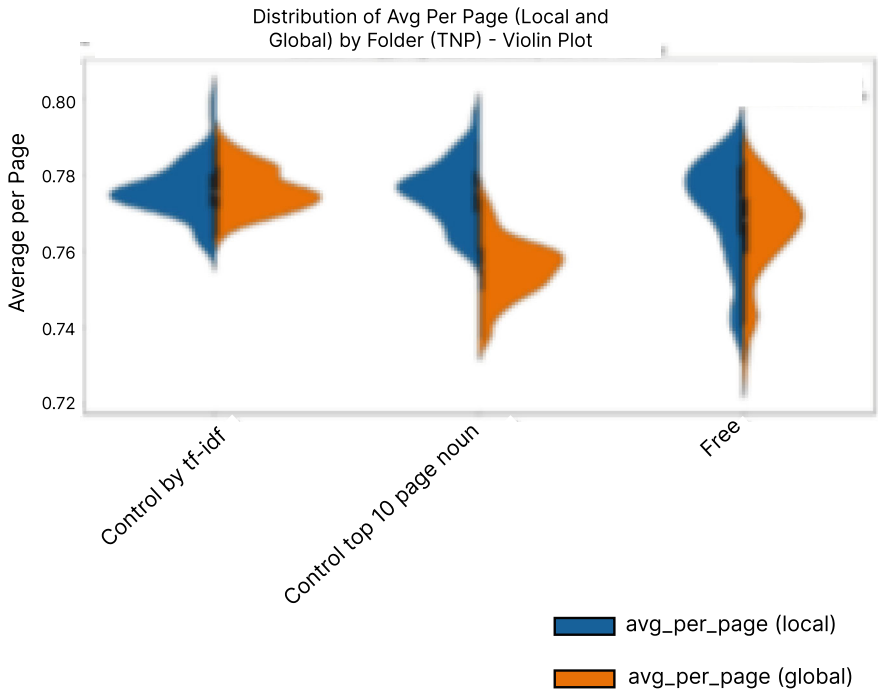


Fig. 8.7 Comparison of context binding performance across three case studies

8.4.4 *Technology Acceptance Model*

The TAM evaluation occurred in two phases to ensure the accuracy of the evaluation results. In the first phase, technical issues arose while running the website, causing some respondents to provide feedback based only on a demonstration presented by the lecturer. This limited the ability to obtain more in-depth feedback on the user experience. In the second phase, the system was fully functional, and all respondents had the opportunity to try it directly. During this phase, they completed a more detailed questionnaire designed to gather specific feedback regarding the user interface, functionality, and their overall experience using the system. A more detailed explanation of this phase is provided in the following subsections.

8.4.4.1 Initial TAM Evaluation

The initial assessment measured PU and PEU using an adapted TAM questionnaire for chatbot-based learning systems [6]. Conducted on November 27, 2024, at Toyo University, it involved 41 Advanced Multimedia students working with a generative AI-themed document content in Japanese. Students selected a preferred language (Japanese, English, or Indonesian), studied a PDF-based lesson, and responded to AI-generated questions, with additional questions provided if they did not reach the 70% threshold.

After the session, participants completed a TAM survey assessing PU, PEU, risk perception (R), and privacy/security concerns (KP). The system was considered effective if it supported generative AI learning and was seen as accessible and easy to use. The study also aimed to identify challenges such as missing key concepts or discomfort during interaction, and examined how background variables and prior AI experience affected overall technology acceptance (Table 8.2).

The evaluation showed a strong correlation (0.991) between PEU and PU, highlighting the importance of usability in shaping users' perceptions of system value (Fig. 8.8). Among PEU indicators, ease of access (PEU2) and understanding questions (PEU1) had the largest contributions, while comfort in system interaction (PEU3) was less influential. On the PU side, delivering key concepts (PU3) and assessment accuracy (PU7) were the most significant factors, whereas understanding documents (PU1) contributed the least. These findings suggest that functional clarity and feedback accuracy are more crucial to user acceptance than comfort or overall content comprehension.

These findings also imply that, to enhance PEU, the system should ensure better accessibility (PEU2) and present easily understandable questions (PEU1). To improve PU, the system should highlight its ability to deliver key concepts (PU3) and provide accurate assessments (PU7). Enhancing comfort in system interactions could involve redesigning the interface or chatbot responses (PEU3). Additionally, the system's ability to aid in understanding documents (PU1) needs to be strengthened.

Table 8.2 List of questions for initial TAM evaluation

No	Question	Type	Category
1	Academic level	Multiple choices	–
2	Are you actively using generative artificial intelligence (like ChatGPT)?	Multiple choices	–
3	This system provides easy to understand questions	Rating scale 1–5	PEU1
4	This system can help to understand the text of a particular document	Rating scale 1–5	PU1
5	This system can provide keywords that you need to learn	Rating scale 1–5	PU2
6	This system allows easy access	Rating scale 1–5	PEU2
7	It is comfortable to interact through a chatbot model	Rating scale 1–5	PEU3
8	The evaluation results you obtain can be easily found	Rating scale 1–5	PEU4
9	This system helps to provide key concepts in the material	Rating scale 1–5	PU3
10	The assessments given by the system help you understand the material	Rating scale 1–5	PU4
11	This system can speed up the course learning process	Rating scale 1–5	PU5
12	This system risks missing important concepts in the material	Rating scale 1–5	R1
13	This system provides security for stored answers	Rating scale 1–5	KP1
14	This system provides security for stored assessment results	Rating scale 1–5	KP2
15	This system provides privacy guarantees for stored answers	Rating scale 1–5	KP3
16	This system provides solutions related to automatic question creation	Rating scale 1–5	PU6
17	This system provides an assessment that is in accordance with the answers given	Rating scale 1–5	PU7
18	Let us know what you think; any feedback is greatly appreciated	Open question	–

8.4.4.2 Second Phase TAM Evaluation

The second evaluation phase occurred in a Machine Learning course at Maranatha Christian University during the winter semester of 2024/2025 academic year. Students answered questions based on Chapters 5 (Support Vector Machines) and 6 (Decision Trees) from the course textbook [15]. Between April 3 and 7, 2025, 46 students independently tested the system using the questions listed in Table 8.3, with 44 achieving the 70% threshold on their first attempt and two on their second.

The success threshold was set at 70%. Each student received one question from a pool of three. If they did not meet the threshold on the first question, the subsequent questions would be presented one at a time. If they still did not succeed after using all available questions, a final question was generated by OpenAI. In this phase (see Fig. 8.9), the system had been improved based on feedback from the first phase. The correlation between PEU and PU remained strong at 0.850, indicating that the easier a system is to use, the more beneficial it feels to users.

In the improved second phase, the system used an adaptive questioning approach with a 70% success threshold, offering up to three predefined questions and a final OpenAI-generated fallback if needed (Fig. 8.9). The correlation between PEU and PU remained strong at 0.850, reinforcing that ease of use increases perceived value. Top PEU contributors included ease of understanding questions (0.710) and interface accessibility (0.608), while identifying AI-generated questions scored lowest (0.415), indicating a need for clearer labels. On the PU side, automatic feedback (0.837), learning efficiency (0.796), and accurate assessments (0.750) were the most influential, whereas keyword provision (0.462) was the least impactful, pointing to potential areas for future improvement, especially in supporting user-driven topic selection.

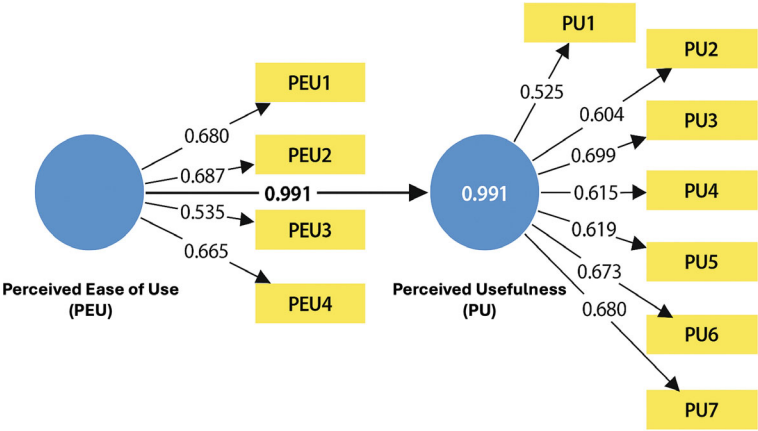


Fig. 8.8 Initial TAM analysis results

Table 8.3 List of questions for the second phase of TAM evaluation

No	Question	Type	Category
1	Does this system help you understand the learning material better?	Rating scale 1–5	PU1
2	Does this system make it easier for you to answer questions or issues?	Rating scale 1–5	PEU1
3	Is the automatic feedback feature helpful in improving your understanding?	Rating scale 1–5	PU2
4	How much does this system help save your study time?	Rating scale 1–5	PU3
5	Is the system interface easy to understand?	Rating scale 1–5	PEU2
6	Are the guides or instructions in the system clear and easy to follow?	Rating scale 1–5	PEU3
7	How comfortable are you using this system in the learning process?	Rating scale 1–5	PEU4
8	How satisfied are you with this system overall?	Rating scale 1–5	PU4
9	Do you feel more involved in the learning process with this system?	Rating scale 1–5	PU5
10	Do you feel motivated to learn more using this system?	Rating scale 1–5	PU6
11	Would you be willing to use this system again in the future?	Rating scale 1–5	PU7
12	Would you recommend this system to others?	Rating scale 1–5	PU8
13	Does this system provide easy to understand questions?	Rating scale 1–5	PEU5
14	Does the question look like it was created by OpenAI?	Rating scale 1–5	PEU6
15	Can this system provide the keywords you need to learn?	Rating scale 1–5	PU9
16	Does this system provide an assessment that is appropriate to the answers given?	Rating scale 1–5	PU10
17	Suggestions and feedback regarding the system.	Open question	(Feedback)

8.4.4.3 Overall Survey Results

Findings from both survey phases consistently highlight that accessibility and interface clarity are crucial for PEU, while features that improve learning efficiency and assessment accuracy greatly influence PU. Participants pinpointed areas for enhancement, including question discoverability and keyword delivery, emphasizing the importance of user-centered design in educational tools.

A comparative analysis reveals significant improvements in system effectiveness between the two phases. Initially, only 5 out of 21 participants scored above 70%, mainly due to technical issues and discomfort with chatbot-style interaction. After system upgrades, 24 out of 26 users achieved the success threshold on their first try. Although the correlation between PEU and PU decreased slightly—from 0.991 to 0.850—overall survey indicators improved: accessibility increased from 0.535 to 0.815, and user understanding from 0.525 to 0.75, demonstrating the system’s improved usability and educational effectiveness.

8.5 Conclusion

The integration of AI into web-based learning platforms offers substantial benefits for scalable education. The proposed system employs LLM to automate question generation and evaluation, enhancing instructional efficiency and reinforcing foundational learning through alignment with Bloom’s Taxonomy. Evaluation results show that

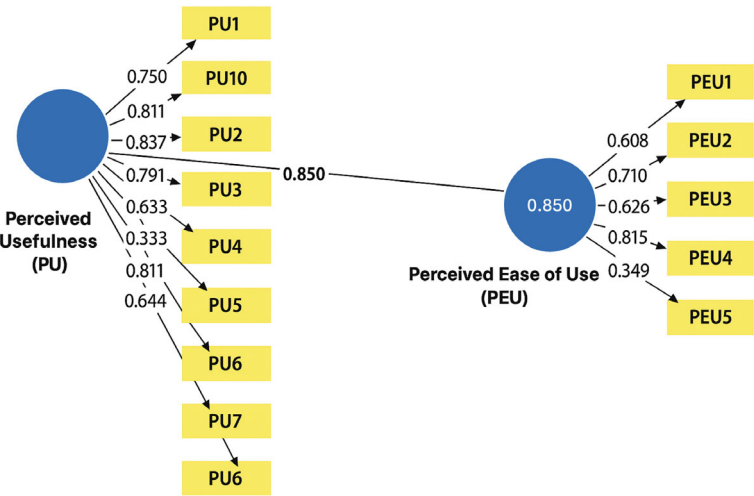


Fig. 8.9 Second phase TAM analysis results

the system maintains student engagement and provides timely, automated feedback, to support comprehension.

Despite its strengths, the system faces challenges related to the variability of LLM-generated outputs, particularly in maintaining consistent question structure. Moreover, its current design is limited in addressing higher-order cognitive tasks such as analysis and synthesis. Future improvements will focus on refining context-anchoring methods, improving cross-language translation accuracy, and adding collaborative learning features. These developments aim to advance the system into a more adaptive and personalized learning solution for diverse educational contexts.

8.6 Implication, Limitation, and Future Work

Experimental results demonstrate the potential of using LLMs to boost student engagement through self-learning. Starting class sessions with automated online tutoring could help, allowing instructors to focus on material they find difficult during in-person classes. University policymakers could also more easily monitor learning progress with outcome-based education.

From an ethical perspective, students should be encouraged to critically evaluate outputs from LLMs. The system presented enables learners to see AI-generated content in action while highlighting the importance of assessing its accuracy. By answering AI-generated questions with textbooks or relevant sources, students build judgment and are guided to operate within academically appropriate limits.

Future development should focus on improving question generation by integrating more advanced AI models that can produce questions with varied difficulty levels and align with specific cognitive taxonomies (e.g., analysis, synthesis). Enhancing answer evaluation—using metrics like BLEU, ROUGE, or logic-based (qualitative) scoring—would enable more thorough assessment of complex responses. Additional features such as visualizations of student performance and personalized feedback would help instructors track progress and tailor instruction.

To support scalability and wider adoption, the system should be tested on various devices and networks. Adding interactive features like discussion forums and instructor-reviewed grading is suggested. Improving backend infrastructure and security, expanding multilingual support, and comparing performance with models like Claude or Gemini can further boost accuracy and content relevance, especially in translated or non-native language situations.

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